

Dog Diving Capstone

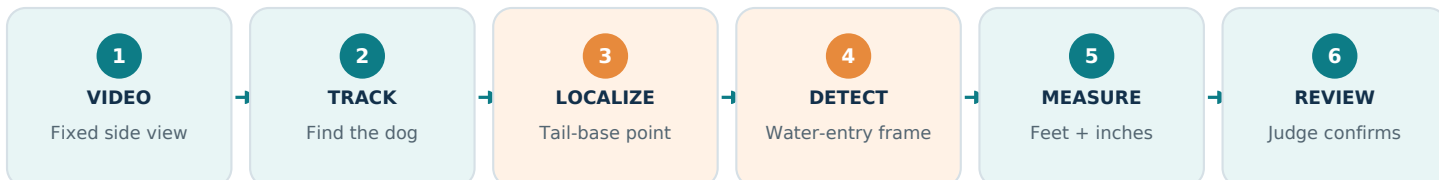
STUDENT HANDOUT

From competition video to a reliable, reviewable jump-distance estimate Technical advisor: Dr. Matthew Korban

PROJECT OBJECTIVE

Design and test an alpha-stage vision system that estimates a dog's jump distance from video and presents the evidence to a human judge.

1 Proposed system



RULE ASSUMPTION - CONFIRM WITH THE CLIENT

For DockDogs Big Air, distance is typically measured from the dock edge to where the base of the tail first breaks the water surface. The client's organization may use a different rule.

2 Minimum viable product

- One distance-jumping discipline
- One fixed, side-facing camera setup
- Offline or near-real-time video processing
- Automatic distance estimate with confidence
- Contact frame and marked measurement point
- Human correction for uncertain cases
- Tested performance against expert scores

3 Technical workstreams

ML

Dog detection, tracking, and tail-base localization

VIDEO

Find the first valid water-contact frame

CV

Dock edge, waterline, and camera calibration

SYSTEM

Convert pixels to distance and report uncertainty

UI

Replay, overlay, score display, and judge correction

4 Success criteria

DISTANCE ERROR

Mean and median absolute error in inches

FRAME ERROR

Difference from the expert-selected contact frame

EFFICIENCY

Processing time per jump or frames per second

TOLERANCE

Percent of jumps within +/-2, +/-4, and +/-6 inches

RELIABILITY

Failure rate and number of cases requiring correction

GENERALIZATION

Results across dogs, distances, lighting, and splash levels

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Execution plan, team ownership, and advising rhythm

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5 Semester roadmap

WEEKS 1-2 Define the problem

Rules, client needs, data inventory, success metrics

WEEKS 3-4 Build the baseline

Calibration plus manual contact-point measurement

WEEKS 5-7 Develop perception

Detection, tracking, labels, and tail-base localization

WEEKS 8-10 Detect water entry

Temporal logic/model and confidence estimation

WEEKS 11-12 Integrate the prototype

End-to-end scoring and review interface

WEEKS 13-14 Test and improve

Error analysis across dogs and conditions

FINAL Demonstrate

Client demo, poster, presentation, and final report

6 Decisions needed first

- Which official scoring rule applies?
- What accuracy will the client accept?
- Is scoring live, offline, or both?
- What videos and official scores exist?
- Will the camera position be controlled?
- What may be stored, shared, and shown?

7 Team ownership

- A** Calibration + distance conversion
- B** Tracking + contact detection
- C** Interface + integration + testing

8 Meeting and reporting expectations

RECOMMENDED CADENCE

- First month: weekly, 30 minutes
- Middle: every two weeks, 30-45 minutes
- Final month: weekly as integration requires

SEND BEFORE EACH MEETING

- Work completed and current evidence
- Problems encountered and attempts made
- Next steps and specific advisor questions

NEXT MILESTONE

Bring a one-page specification: official rule, client requirements, available data, camera setup, quantitative target, role assignments, and first baseline plan.